

# Nathan Shoults

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[Nathan-Shoults.github.io/portfolio](https://Nathan-Shoults.github.io/portfolio)



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## EDUCATION

### **Northern Arizona University — B.S. in Electrical Engineering**

Aug 2021 - May 2024

Directed a multidisciplinary team of three graduate Mechanical Engineers and two Electrical Engineers in the design, construction, and implementation of a Scanning Tunneling Microscope. Achieved atomic resolution microscopy for quantum dot research within the School of Informatics, Computing, and Cyber Systems at Northern Arizona University. Reorganized and worked with the College of Engineering, Informatics, and Applied Sciences to allocate a space for the IEEE club.

### **Santa Rosa Junior College — General Education**

Aug 2020 - May 2021

## EXPERIENCE

### **Carrington Construction — Carpenter, Electrician, and aided the design team with CAD expertise**

September 2024 - Present

Contributed to the construction of a legacy home at 633 Quarry Hill in Sonoma, gaining an in-depth understanding of the processes involved in building both commercial and residential real estate.

### **IEEE Club at NAU — Vice President**

Dec 2023 - May 2024

Assisted in the planning and fundraising for competitive events for the following year such as the Arizona Robotics League.

Tutored first and second year EE students in classes including but not limited to: Calc 3, Differential Equations, EE 188, EE 101, EE 385, EE 404

### **McMillan Bar and Kitchen — manager, bartender, server, bouncer, host, and dishwasher**

Aug 2022 - May 2024

Gained an in-depth understanding of what is required to run a restaurant business. Was the sole maintenance personnel for the oldest standing building in flagstaff (1886).

## SKILLS

Experienced with a plethora of EDA platforms such as Kicad, Altium, and Fusion 360 in PCB design and fabrication. Most recent project: a constant current supply for a 50 W handheld IR laser.

Experienced in the following programming languages: C, C#, C++, python, html, CSS, G-code, Basic, HLA, Java, Microcode, OpenCL, simulink, Matlab, and many others.

Experienced with oscilloscopes, digital multimeters, function generators, Arduino, Raspberry Pi, and a plethora of microcontrollers and SBC's.

Proficient in the setup, calibration, and troubleshooting of advanced lab equipment such as signal analyzers, spectrum analyzers, and logic analyzers, ensuring accurate data collection for experimental and research purposes.

Seven years experience with Fusion 360, SolidWorks, and SLS 3D printing

## AWARDS

**Graduated with Honors** from Northern Arizona University

**Doyle Scholarship** during enrollment at SRJC